



Amir Mahdi Amani

Robotics & AI Engineer  
ROS/ROS2 | Computer Vision

- Location:** Cologne, Germany
- Bernhardstr. 119, 50968 Cologne
- +49-152-11555803
- amirmahdi.amani7th@gmail.com
- Website:** am-amani.github.io
- GitHub:** am-amani

Focus Areas

- Robotics Software & Systems Integration
- Autonomous Navigation & Perception
- Image Processing
- Robot Learning & Deep Reinforcement Learning
- Simulation, Testing & Automation

Skills

Programming & Scripting:

Python

C++

SQL

Bash

MATLAB

AI & Computer Vision:

PyTorch, TensorFlow

OpenCV

Codex, Claude Code

Robotics:

ROS / ROS2

Gazebo

MoveIt

SLAM, AMCL

Professional Summary

Robotics and AI engineer with hands-on experience in ROS/ROS2, Python/C++, autonomous navigation, computer vision, deep learning, robot testing, and simulation. Built and evaluated multi-sensor learning-based navigation systems in ROS/Gazebo, configured and tested mobile robot platforms, and developed ROS/MoveIt/RGB-D solutions in robotics research. Current industry work adds production-focused data analysis, automation-oriented workflows, validation, and technical documentation. Strong fit for roles combining robotics software, AI integration, simulation, system testing, and deployment.

Professional Experience

- Oct. 2024 – present

**AI Solution Engineer – Network Planning, Optimization & Automation Specialist**

Deutsche Netzplanung GmbH, Cologne, Germany

Develop and support data-driven planning and automation-oriented workflows for telecommunications infrastructure. Use QGIS and geospatial data to analyze and validate project information, classify deployment regions, troubleshoot planning issues, and support optimization activities. Collaborate with engineers and translate technical requirements into practical workflows, documentation, reports, and stakeholder-ready outputs.
- Mar. 2024 – Sep. 2024

**Fiber Network Planning and Optimization (Part-Time)**

Deutsche Netzplanung GmbH, Cologne, Germany

Supported network-planning projects and the optimization of fiber infrastructure. Used QGIS for surface-area analysis and regional classification, reviewed technical data, assisted engineers with planning decisions, and supported implementation and project documentation.
- Feb. 2024 – Sep. 2024

**Robotic Research Assistant (HiWi)**

Humanoid Robots Lab, University of Bonn

Configured, tested, and documented ROS-based navigation capabilities for Unitree GO1 and TurtleBot2 platforms, including SLAM and AMCL. Diagnosed operational issues, evaluated different robot modes, upgraded localization/navigation/mapping workflows, and prepared practical setup and usage documentation for researchers.
- Sep. 2019 – Nov. 2019

**Research Internship – Design-to-Robotic-Production and Operation**

TU Delft, The Netherlands

Developed and integrated ROS-based robotics solutions using MoveIt and Kinect V2. Applied forward/inverse kinematics and RGB-D perception for calibration, filtering, segmentation, clustering, object detection, and pose estimation, with practical exposure to KUKA KR 10 industrial-robot workflows and controller integration.

Publication

Journals

- Amani, A. M.; Amani, S.; MajidiRad, A. A Unified Conditional Policy for Multi-Robot Navigation via LiDAR-to-Vision Distillation. *Machines* **2026**, *14*, 936. DOI.

Multi-Robot Navigation

Policy Distillation

LiDAR-to-Vision

Deep RL

Selected Robotics & AI Projects

- 2023 – 2024

**Master Thesis – Multi-Sensor Robotic Navigation**

Robot Learning / Deep RL

Developed and evaluated a teacher-student navigation framework for Pioneer 3-DX and TurtleBot3 Waffle using TD3, LiDAR, vision, and ROS/Gazebo. Trained learning-based navigation policies, tested multiple robot/sensor configurations, and studied policy transfer between sensing modalities.

ROS/Gazebo

TD3

LiDAR/Vision

Policy Distillation

## Skills (cont.)

### Tools & Platforms:

Linux, Docker

Git, CMake

QGIS

## Short Bio

AI and robotics engineer with an M.Sc. in Computer Engineering, specializing in Artificial Intelligence and Robotics. Experienced in Python, C++, deep learning, computer vision, autonomous navigation, geospatial analysis, technical testing, and documentation. Research work includes multi-sensor robot navigation using teacher-student policy distillation in ROS/Gazebo. Currently developing data-driven solutions for network-planning and automation workflows in Germany.

## Languages

English (Professional)

German (B1)

## Personal

Interests include cooking, karaoke, dancing, reading, gym training, open discussions, and films. As a member of the Computer Science Association, helped organize an annual ACM programming competition for students from different universities in South Khorasan and conducted monthly C++ and Python workshops for freshmen. Also organized weekly English discussion sessions, special events, and Persian karaoke activities. Later created an online page for Persian karaoke content and community engagement.

## Selected Robotics & AI Projects (cont.)

Dec. 2023

**Real-Time Camera Pose and Tumor Detection for Endoscopy** Computer Vision

Combined camera-motion tracking with HSV filtering, Canny edge detection, Hough Circle Transform, background subtraction, and morphological operations for real-time pose estimation and potential-tumor detection.

OpenCV

Pose Estimation

Image Processing

Detection

Oct. 2023

**TIAGO Mobile Robot Navigation and Obstacle Detection** Robotics

Implemented ROS navigation from a user-defined start to a destination and developed an Action Client/Server system for detecting and reporting movable obstacles from laser-sensor data.

ROS

Navigation

Laser Sensing

ActionLib

Jun. 2023

**Image Colorization with Conditional WGAN** Deep Learning

Built a PyTorch conditional WGAN with a U-Net architecture and skip connections, trained on more than 50,000 grayscale images, and used conditional inputs and image-processing techniques for controlled colorization.

PyTorch

WGAN

U-Net

Image Colorization

May 2023

**TinyPointNet for 3D Local Descriptors** 3D Data Processing

Designed a compact PointNet variant for efficient local feature extraction from point clouds and evaluated it on diverse 3D datasets and benchmark tasks.

PointNet

Point Clouds

PyTorch

3D Descriptors

Jun. 2022

**Human Hands Detection and Segmentation** Computer Vision

Trained a custom YOLOv5 model with PyTorch and implemented C++ segmentation using contours, thresholding, and morphological operations.

YOLOv5

PyTorch

C++

Segmentation

## Education

### Postgraduate Studies

Sep. 2021 – Sep. 2024

**M.Sc. in Computer Engineering – Artificial Intelligence & Robotics** University of Padova, Italy

**Thesis:** Multi-Sensor Robotic Navigation Using a Teacher-Student Framework.

**Grade:** 96/110 (approximately 1.9/4).

AI & Robotics

Machine Learning

Computer Vision

3D Data

Dec. 2023 – Sep. 2024

**Erasmus+ Exchange – Thesis and Research Internship** University of Bonn, Germany

Research at the Humanoid Robots Lab under Prof. Emanuele Menegatti, Prof. Maren Bennewitz, and Murad Dawood.

ROS/Gazebo

TD3

LiDAR/Vision

Robot Navigation

### Undergraduate Study

Sep. 2015 – Jun. 2020

**B.Sc. in Computer Software Engineering** University of Birjand, Iran

Coursework in C++, data structures, algorithm design, robotics, and image processing. **Grade:** 14.93/20.

C++

Algorithms

Robotics

Image Processing

## Other Training

2019 – 2021

### Selected AI and Machine-Vision Certificates

DeepLearning.AI / Stemmer Imaging

TensorFlow for AI, machine learning and deep learning (2021); machine-vision solutions with HALCON and MERLIC (2019).

TensorFlow

Machine Learning

HALCON

MERLIC

## Honours and Awards

Erasmus+ Traineeship Scholarship  
Regione Veneto Scholarship (2.5 years)

University of Padova

University of Padova

## Conference Experience

Mar. 2023

### Speaker – 15th International Petroleum Technology Conference

Bangkok, Thailand

Presented “Geological CO2 Leak Monitoring Using Joint Seismic-CSEM Inversion Based on CRIM and White Equations” on behalf of a peer and communicated the technical content to an international audience.

## Teaching Experience

Teaching Assistant – Differential Equations  
Teaching Assistant – General Mathematics I

University of Birjand

University of Birjand

## References

Ref. 1

**Prof. AmirHossein MajidiRad**  
amajidir@pnw.edu

Purdue University Northwest

Ref. 2

**Prof. Emanuele Menegatti**  
emanuele.menegatti@unipd.it

University of Padova